

Project Report: BANK CUSTOMER SEGMENTATION USING MACHINE LEARNING

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1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

1.1 Project Overview

This project implements an end-to-end **Unsupervised Machine Learning system** to segment banking customers based on transactional behavior. Processing **1,500 customer records across 14 numerical financial attributes**, the system executes data cleaning, normalization, optimal cluster determination, multiple clustering algorithms, validation, and persona generation to support targeted financial strategies.

1.2 Business Problem

Traditional generic customer classifications in banking lead to three major operational issues:

- **Misaligned Product Marketing:** Premium products offered to low-activity users, while active installment shoppers lack suitable BNPL offers.
 - **Unidentified Risk Exposure:** High cash-advance reliance and multiple active loans go unnoticed, delaying risk intervention.
 - **Suboptimal Customer Retention:** High-value spenders receive default services rather than premium engagement, increasing churn risk.
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1.2 Business Problem

Traditional banking systems frequently rely on generic customer classifications and one-size-fits-all marketing strategies.

This creates several operational problems:

Misaligned Product Marketing

Premium rewards and investment products may be promoted to low-activity customers, while frequent installment-based shoppers may not receive suitable installment or BNPL offerings.

Unidentified Risk Exposure

Customers who depend heavily on cash advances and maintain multiple active loans may not be identified early enough for financial risk intervention.

Suboptimal Customer Retention

High-value customers with substantial balances, high credit limits, and frequent purchases may not receive appropriate premium services, reducing customer engagement and retention.

1.3 Project Objectives

1. Load, inspect, and preprocess raw financial behavioral data.
2. Impute missing values and standardize features for distance calculations.
3. Determine and validate the optimal cluster count scientifically.
4. Execute K-Means and Hierarchical Clustering, verifying stability and agreement.
5. Profile customer segments to map actionable, differentiated banking products.

1.4 Solution Overview

Pipeline Flow: Raw Dataset → Data Cleaning → Median Imputation → StandardScaler → Optimal K Selection → K-Means + Hierarchical Clustering → PCA → Customer Profiling → Product Strategy

2. DATASET & PREPROCESSING

2.1 Dataset Overview

The dataset contains **1,500 customer records** with **15 columns** (1 customer identifier and 14 numerical financial features), adhering to the Kaggle Credit Card Dataset for Clustering specifications.

2.2 Feature Definitions

Feature	Purpose	Missing Values
CUST_ID	Unique customer identifier (Excluded from ML)	0
BALANCE	Account balance maintained by customer	45 (3.0%)
PURCHASES	Total amount of purchases made from account	0
ONEOFF_PURCHASES	Maximum purchase amount done in one-go	0
INSTALLMENTS_PURCHASES	Amount of purchase done in installment	0
CASH_ADVANCE	Cash in advance given by the user	0
PURCHASES_FREQUENCY	Frequency of purchases (0 to 1)	0
PURCHASES_TRX	Number of purchase transactions made	0
CREDIT_LIMIT	Limit of credit card for user	22 (1.5%)
PAYMENTS	Amount of Payment done by user	0
MINIMUM_PAYMENTS	Minimum amount of payments made by user	60 (4.0%)
PRC_FULL_PAYMENT	Percent of full payment paid by user	0
LOAN_HOLDINGS	Number of active loan holdings	0
TENURE	Tenure of credit card service for user	0

2.3 Missing Value Handling

Missing values were detected in BALANCE (45), CREDIT_LIMIT (22), and MINIMUM_PAYMENTS (60). Because financial attributes exhibit strong right-skewness due to high-spending outliers, mean imputation would introduce bias. Median imputation was selected to preserve data distribution stability using the formula:

$$\text{Missing}(X_j) = \text{Median}(X_j)$$

The imputed median values were: **BALANCE = \$2,361.41**, **CREDIT_LIMIT = \$6,162.25**, and **MINIMUM_PAYMENTS = \$656.88**.

2.4 Feature Scaling

Distance-based algorithms like K-Means are sensitive to feature scales. Without scaling, large magnitude attributes (e.g., CREDIT_LIMIT) dominate small ratio features (e.g., PURCHASES_FREQUENCY). CUST_ID was excluded, and the remaining 14 features were standardized using StandardScaler via z-score normalization:

$$z = (x - \mu) / \sigma$$

3. MACHINE LEARNING METHODOLOGY

3.1 Overall ML Pipeline

System Architecture: Dataset → Preprocessing → Scaling → Optimal Cluster Selection → K = 4 → (K-Means + Hierarchical Clustering) → PCA Validation → Cluster Profiling

3.2 Optimal Cluster Selection

The cluster count was determined using three independent validation techniques rather than arbitrary selection:

- **Elbow Method (WCSS):** Evaluates Within-Cluster Sum of Squares across K=1 to 10. WCSS dropped from 21,000.0 at K=1 to 5,120.7 at K=4 (a 75.6% reduction), with the elbow distinctly forming at K=4 before flattening.

Silhouette Score: Measures cohesive clustering and cluster separation. Scores evaluated as follows:

K	Silhouette Score
2	0.3917
3	0.4699
4	0.4846
5	0.3984

- **Ward-Linkage Dendrogram:** Truncates into 4 major structural branches at distance threshold $d \approx 35$.

Conclusion: All three validation techniques support K=4; therefore, four customer segments were selected.

3.3 K-Means Clustering

K-Means partitions customers by minimizing squared Euclidean distance to learned centroids. The final model implementation uses `KMeans(n_clusters=4, random_state=42)`. Learned centroids provide a mathematical definition for each cluster, enabling straightforward assignment of new incoming customer records.

3.4 Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering was implemented with Ward linkage as an independent bottom-up validation technique, minimizing error sum-of-squares variance during cluster merges.

3.5 Clustering Validation & Stability

To evaluate consistency across algorithms, the Adjusted Rand Index (ARI) was calculated between K-Means and Hierarchical Clustering, yielding **ARI = 1.000** (complete agreement). Furthermore, K-Means was tested across 5 random initializations (`random_state = 42, 100, 2024, 777, 999`), obtaining an **average ARI = 1.000**, proving total cluster assignment stability.

3.6 Dimensionality Reduction (PCA)

Principal Component Analysis reduced the 14-dimensional feature space to 2 components for visual validation. **PC1 explained 52.4%** of the variance (driven by purchase volume, payments, credit limits), and **PC2 explained 21.6%** (driven by cash advance usage and minimum payments). Together, PC1 and PC2 capture **74.0% of total dataset variance**, showing well-separated cluster regions in 2D space.

4. CUSTOMER SEGMENTATION RESULTS

4.1 Customer Distribution

Cluster	Customers	Percentage	Segment Persona
Cluster 0	525	35%	Conservative Savers / Low Activity
Cluster 1	225	15%	Installment Budgeters & Frequent Spenders
Cluster 2	450	30%	High-Balance Power Spenders

Cluster	Customers	Percentage	Segment Persona
Cluster 3	300	20%	Cash-Advance & Liquidity Seekers

4.2 Discriminative Cluster Profiles

Discriminative Attribute	Cluster 0	Cluster 1	Cluster 2	Cluster 3
BALANCE	\$829.54	\$1,556.48	\$5,451.59	\$4,458.34
PURCHASES	\$250.89	\$2,217.72	\$4,342.34	\$517.21
INSTALLMENTS_PURCHASES	\$124.75	\$1,948.21	\$1,089.42	\$333.24
CASH_ADVANCE	\$110.48	\$158.39	\$314.11	\$4,894.03
PURCHASES_FREQUENCY	0.26	0.85	0.90	0.31
CREDIT_LIMIT	\$3,051.85	\$5,495.20	\$12,082.79	\$7,581.83
LOAN_HOLDINGS	0.21	0.72	0.65	1.80

4.3 Customer Personas

1. Cluster 0 — Conservative Savers / Low Activity (35%)

- **Key Characteristics:** Low balances (~\$830), minimal purchase volume (~\$251/yr across ~5 transactions), and negligible cash advances.
- **Risk Profile:** Low Risk / Low Yield.
- **Recommended Products:** High-Yield Savings Accounts, basic zero-fee credit cards, and automated micro-investment plans.

2. Cluster 1 — Installment Budgeters & Frequent Spenders (15%)

- **Key Characteristics:** High purchase frequency (0.85, ~28 transactions/yr) dominated by installment purchases (~\$1,948 of \$2,218 total).
- **Risk Profile:** Low Risk / High Engagement.
- **Recommended Products:** 0% APR merchant installment offers, cashback rewards cards, and BNPL integrations.

3. Cluster 2 — High-Balance Power Spenders (30%)

- **Key Characteristics:** Substantial balances (~\$5,452), massive purchase volumes (~\$4,342/yr across ~45 transactions), and high credit limits (~\$12,083).

- **Risk Profile:** Low Risk / High Value.
- **Recommended Products:** Platinum rewards credit cards, private wealth management, and high-limit lines of credit.

4. Cluster 3 — Cash-Advance & Liquidity Seekers (20%)

- **Key Characteristics:** Heavy cash advance dependency (~\$4,894), elevated balances (~\$4,458), multiple active loans (1.8 avg), and low direct purchases.
- **Risk Profile:** Moderate-High Credit Risk.
- **Recommended Products:** Debt consolidation loans, low-interest balance transfer cards, and personal financial advisory services.

5. IMPLEMENTATION / NOTEBOOK WORKFLOW

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Cell	Operation	Output
Cell 0	Imports & Environment Setup	Python/Scikit-Learn environment ready
Cell 1	Dataset Loading & Inspection	1,500 × 15 DataFrame loaded
Cell 2	Missing Value Analysis	Null counts identified across features
Cell 3	Median Imputation	Fully imputed clean dataset
Cell 4	Feature Selection & Scaling	Standardized 14 features (z-scores)
Cell 5	WCSS & Silhouette Analysis	Elbow curve (plots/elbow_curve.png)
Cell 6	Final K-Means Execution	Four cluster labels generated
Cell 7	Hierarchical Clustering	Validation (ARI = 1.000)
Cell 8	PCA Transformation	2 Principal Components (74.0% var)
Cell 9	PCA Visualization	Scatter plot (plots/pca_clusters_2d.png)
Cell 10	Profiling & Data Export	data/bank_customer_data_clustered.csv

6. RESULTS & ANALYTICAL FINDINGS

- **How many customer segments exist?** Exactly 4 distinct customer segments exist.
- **Why K=4?** Supported by WCSS elbow point (75.6% drop), peak Silhouette score (0.4846), and Ward dendrogram tree branch truncation.
- **What characterizes each segment?** Cluster 0: Low activity savers; Cluster 1: Frequent installment spenders; Cluster 2: High-balance power buyers; Cluster 3: Cash advance dependency with multiple loans.

- **Does PCA support the segmentation?** Yes, PC1 and PC2 capture 74.0% of total variance and exhibit clear visual cluster separation in 2D space.
- **Are the clustering results stable?** Yes, verified by ARI = 1.000 between algorithms and across five random initializations.
- **Can segmentation support product design?** Yes, each segment directly maps to tailored products (savings, BNPL, premium wealth cards, debt consolidation).

7. LIMITATIONS & REAL-WORLD APPLICABILITY

7.1 Major Limitations

- **Financial Attributes Alone:** The dataset lacks demographic information (age, income, location) and external credit scores (FICO/CIBIL), restricting comprehensive risk modeling.
- **Static Snapshot:** Operates on fixed historical data without capturing dynamic customer lifecycle transitions over time.

7.2 Future Improvements

Production deployment can enhance system capabilities by incorporating real-time transaction streaming, credit bureau API integrations, time-series customer behavioral tracking, automated dynamic segment reassignment, and real-time risk monitoring engines.

8. CONCLUSION

This project successfully implemented an end-to-end unsupervised machine learning pipeline for bank customer segmentation. Combining data preprocessing, median imputation, and feature standardization, the system established an optimal cluster count of $K=4$, validated independently via the Elbow Method, Silhouette Analysis (0.4846), and Ward Dendrograms. Clustering results demonstrated perfect algorithm agreement and initialization stability ($ARI = 1.000$) between K-Means and Hierarchical Clustering, while 2D PCA captured 74.0% of feature variance. The resulting four customer personas (Conservative Savers, Installment Budgeters, Power Spenders, and Liquidity Seekers) provide a robust foundation for targeted, data-driven banking product strategies and automated risk monitoring.